Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Abstract

1 Introduction

1.1 Motivation and research question

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3.1 Spanish wholesale electricity market: a sequential-market overview

3.2 The MTU15 reform sequence

OJO, IF REE RESTRICTS THE RAMPS (QUANTITY CANNOT CHANGE A LOT), THE MTU15 REFORM WILL AFFECT THE BIDS WITH PRICE ADJUSTMENTS INSTEAD OF QUANTITY ADJUSTMENTS, SO THIS AMPLIFIES MARKET POWER EVEN MORE!

3.3 Data sources

3.4 Critical-vs-flat hours: empirical calibration

Our identification strategy relies on the idea that the reform does not give extra strategic space for firms to exploit in all hours (REFERENCE TO OUR MODEL). The reform's strategic effect is concentrated in hours where the *residual demand* (electricity demand minus wind and solar production) varies substantially within the hour — those are the hours where finer time resolution allows firms to set quarter-specific bids that match the within-hour conditions.

Classification criterion. For each clock-hour h, define the within-hour variance of residual demand,

$$\sigma^2_{\text{within}}(h) \;=\; \mathbb{E}_d\left[\tfrac{1}{4}\sum_{q=1}^{4}\big(D_{h,q,d}-\bar{D}_{h,d}\big)^2\right], \qquad D_{h,q,d}=\text{demand}_{h,q,d}-\text{wind}_{h,q,d}-\text{solar}_{h,q,d}, \tag{1}$$

where the expectation is taken across days d and the variance is taken across the four 15-minute quarters q of clock-hour h. **Critical hours** are those with high σ^2_{within} (large within-hour movement in residual demand, in either direction); **flat hours** are those with low σ^2_{within}. Borderline hours (moderate variance, no clean class assignment) are dropped from the analysis.

Applying this criterion to Spain 2025 yields the following grouping:

Pattern	**Hours (local)**	**Mechanism**
Overnight valley	01:00–04:00	Demand low and sta slack
Morning ramp + solar entry	05:00–09:00	Demand rises sharply the hour); solar prod between 06:00 and 09
Midday solar plateau	11:00–15:00	Demand flat at its da peak but variable acro
Afternoon ramp + solar exit	16:00–20:00	Demand rises into th production fades
Post-peak descent	20:00–23:00	Demand falls sharply (1.5–1.9 GW within th gone
Transitional	00:00–01:00, 04:00–05:00, 09:00–10:00, 15:00–16:00, 23:00–24:00	Mixed/small variation

The critical-hour set used in the empirical analysis is 05:00–09:00 and 16:00–23:00 (eleven clock-hours). The flat-hour set is 01:00–04:00 (three clock-hours). The full 24-hour ranking by σ^2_{within}, mean demand, mean solar, and within-hour demand change is reported in Appendix B.5.

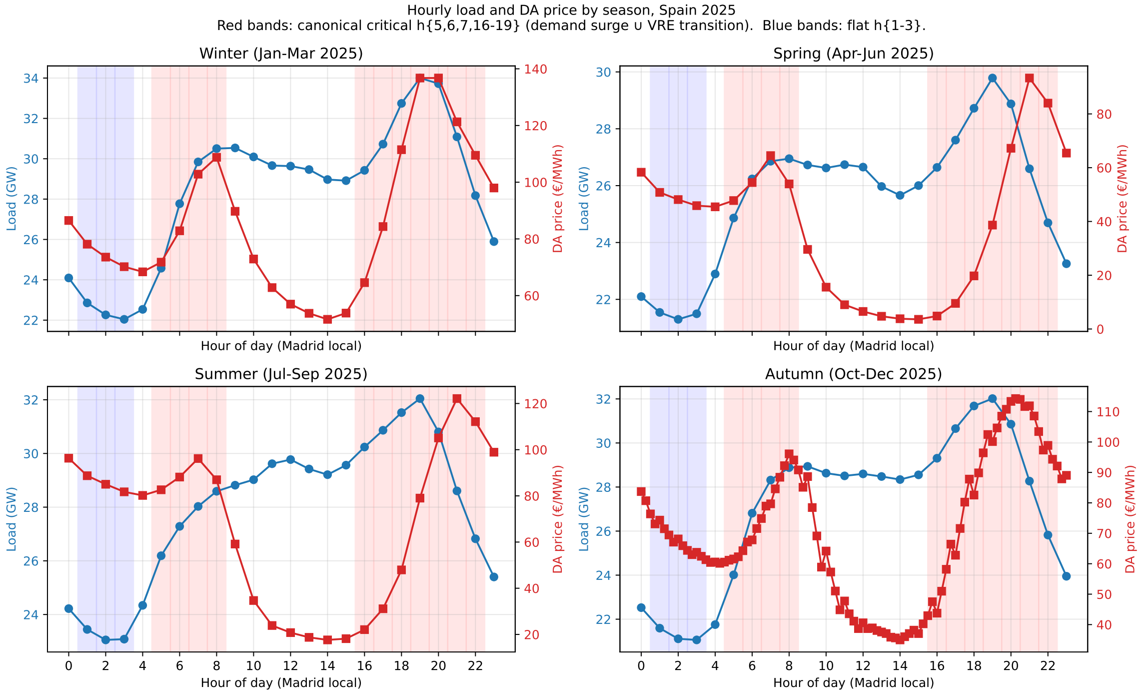

Figure 1: Hourly load and DA price profile, by season, Spain 2025. Four-panel decomposition by season. Blue line: realized Spanish electricity demand; the slope of the line is the within-hour demand ramp rate. Red line: day-ahead clearing price. Red bands mark the *critical hours* (05:00–09:00 and 16:00–23:00, local time): high within-hour variance of residual demand. Blue bands mark the *flat hours* (01:00–04:00): low and stable demand, no solar, system slack. Critical hours combine the morning ramp, the afternoon transition into the evening peak, and the post-peak descent; flat hours are the overnight valley.

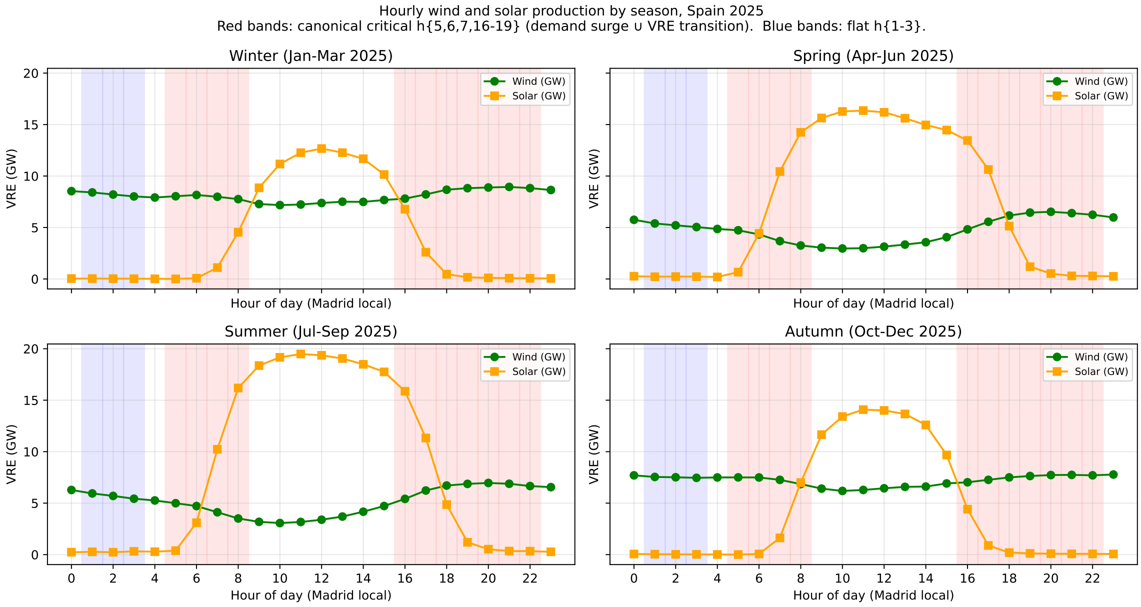

Figure 2: Hourly wind and solar production by season, Spain 2025. Four-panel decomposition by season. Red bands mark the *critical hours* (05:00–09:00 and 16:00–23:00). Blue bands mark the *flat hours* (01:00–04:00). Solar production is concentrated in midday hours and fades sharply at the boundaries of the critical-hour bands; wind is broadly flat across the day with mild seasonality. The combination implies that critical hours systematically coincide with low renewable supply year-round, reinforcing the case for treating them as the strategically relevant time-varying window.

3.5 Identification strategy and treatment-effect set

Table 1: Pivotality rate by parent firm and hour-class, CCGT only, October–December 2025. Pivotality is the unit-level price-setter rate: tranches straddle the DA clearing price (i.e., $n_{\text{below}} > 0$ and $n_{\text{above}} > 0$), evaluated per (date, unit, period) and aggregated by parent firm × hour-class. The treatment in our identification is at the *hour* level (critical vs flat). Firms enter the analysis as subjects: pivotal firms (top panel) are those expected to react to the treatment because they have a strategic margin in critical hours; non-pivotal firms (bottom panel) serve as placebo because they bid at marginal cost regardless of hour. Source: `notebooks/memos/_pivotality_by_firm_critical_hours.md`.

Firm (parent)	flat (01:00–04:00)	critical (05:00–09:00, 16:00–23:00)	Δ crit−flat
Pivotal firms (expected to react to the hour-level treatment)			
EDP-Spain	0.523	0.998	**0.475**
Naturgy	0.018	0.515	**0.497**
EDP-Portugal	0.744	0.865	0.121
Iberdrola	0.009	0.156	0.147
Endesa	0.124	0.240	0.117
Non-pivotal firms (placebo)			
Engie España	0.000	0.002	0.002
Moeve (formerly Cepsa)	0.005	0.005	−0.001
Repsol	0.000	0.000	0.000
TotalEnergies	0.000	0.000	0.000

3.6 Parallel-trends discussion

4 Bidding-behavior evidence of the mechanism

4.1 Per-firm bid-shape patterns at hour-of-day resolution

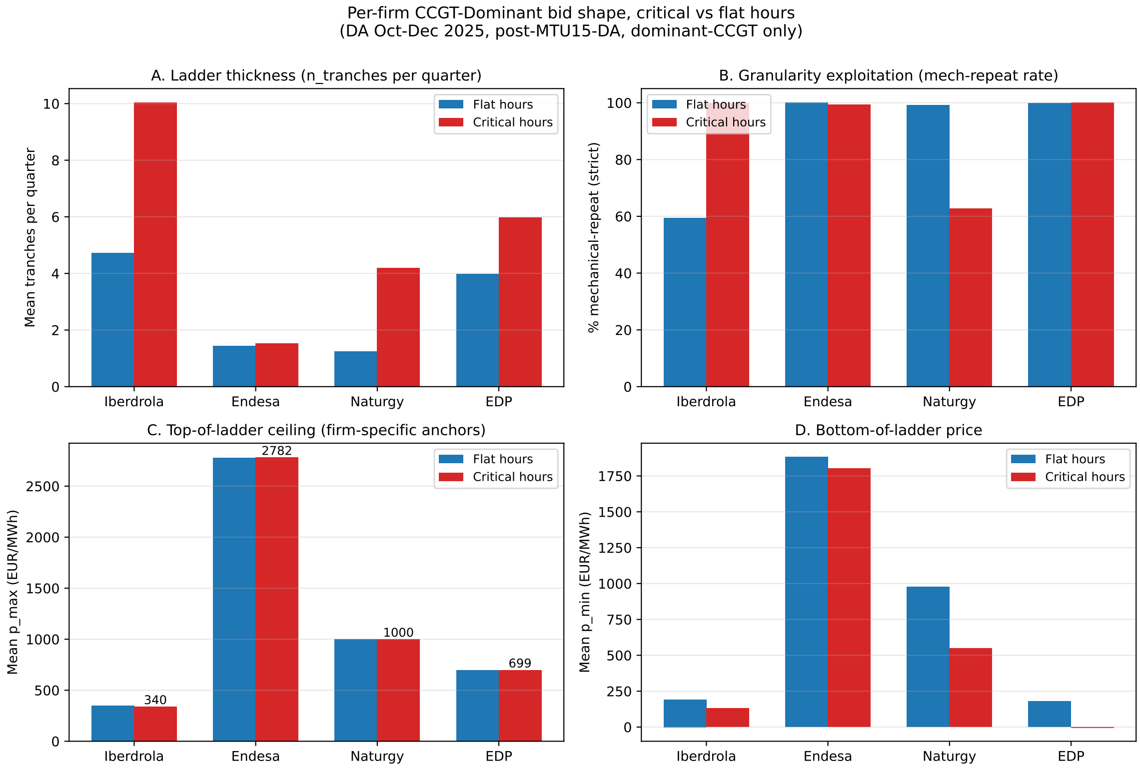

Figure 3: Per-firm CCGT bid-shape patterns, October–December 2025 day-ahead market. Four-panel decomposition of dominant CCGT bidding by firm (Iberdrola, Endesa, Naturgy, EDP-Spain). Panels show per-firm critical vs flat-hour means for tranche count, mechanical-repeat rate, top-of-ladder price, and bottom-of-ladder price. Reveals two distinct strategic types: Iberdrola (uniform enricher, richest ladders, no quarter variation) and Naturgy (granularity exploiter, varies bids quarter-by-quarter). Endesa and EDP-Spain show mid-strategic and inactive patterns respectively. Source: `scripts/analysis/firm/per_firm_hourly_ccgt_bidshape.py`.

4.2 Granularity exploitation: ladder, within-hour quarter variation, and their interaction

4.3 Pre- vs post-MTU15-DA structural changes

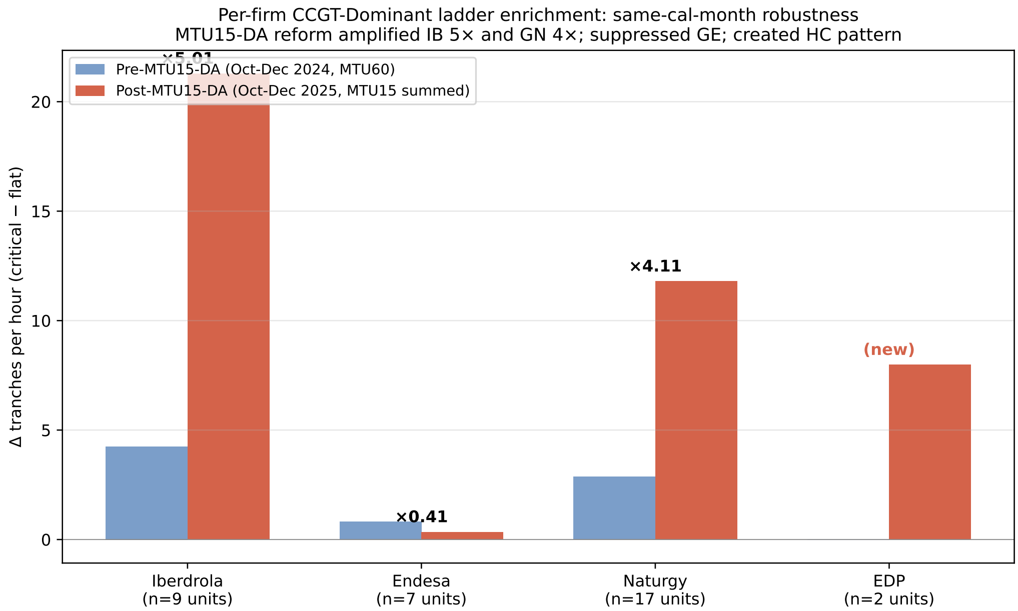

Figure 4: Pre- vs post-MTU15-DA amplification of the critical–flat differential, by firm. Same-calendar-month comparison (October–December 2024 vs October–December 2025) of CCGT bid-shape critical-flat differential by firm. Iberdrola amplifies 5.0×, Naturgy 4.1×, EDP-Spain creates a positive differential where there was none, while Endesa's pattern is suppressed. Indicates asymmetric strategic response to the granularity reform across the dominant tier. Source: `scripts/analysis/bid/per_firm_pre_vs_post_mtu15da.py`.

4.4 Heterogeneity by firm and technology

5 Main empirical results

5.1 Within-day DiD on q_2: primary outcome

Table 2: B1 – Within-day DiD on q_2 (Ito–Reguant strategic IDA upward adjustment). Outcome is q_2 in MWh per (unit, clock-hour), defined as the sum of `pibci` across IDA sessions converted to MWh and aggregated to clock-hour resolution. Specification: $q_{2,fdh} = \alpha + \beta_1 \texttt{crit}_h + \beta_2 \texttt{post}_d + \beta_3(\texttt{crit}_h \times \texttt{post}_d) + \gamma_f + \delta_{\text{DOW}(d)} + \varepsilon_{fdh}$. Firm and day-of-week fixed effects; cluster-robust standard errors by date. Sample: same-calendar-month windows, October–December 2024 and October–December 2025 (both windows post-MTU15-IDA reform of March 2025). Critical hours: $h \in \{5, 6, 7, 8, 16, 17, 18, 19, 20, 21, 22\}$ (canonical: demand surge $\cup$ VRE transition; excludes post-peak h{20-22} which are descending and h4 which is too early for system tightness). Flat hours: $h \in \{1, 2, 3\}$. Treatment group: firms with $\geq 10\%$ pivotality in critical hours (Iberdrola, Endesa, Naturgy, EDP-Spain, EDP-Portugal). Placebo: firms with $\sim 0\%$ pivotality (Repsol, Engie España, TotalEnergies, Moeve). Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; headline interpretation relies on *** given the large effective sample size.

	All firms (1)	Treatment group (2)	Placebo group (3)
β_3: `crit` $\times$ `post`	0.10	5.69***	-3.02^{***}
cluster-robust SE	(0.09)	(1.08)	(0.31)
p	0.2756	0.0000	0.0000
β_1: `crit`	-0.10	-5.97	3.29
β_2: `post`	-0.12	-14.49	0.14
$\bar{y}$ (pre, flat)	0.12	16.51	-0.04
$\bar{y}$ (pre, crit)	-0.05	10.45	3.29
$\bar{y}$ (post, flat)	0.00	1.84	0.20
$\bar{y}$ (post, crit)	-0.05	1.55	0.52
n	4,581,582	317,686	568,036
clusters G (dates)	184	184	184

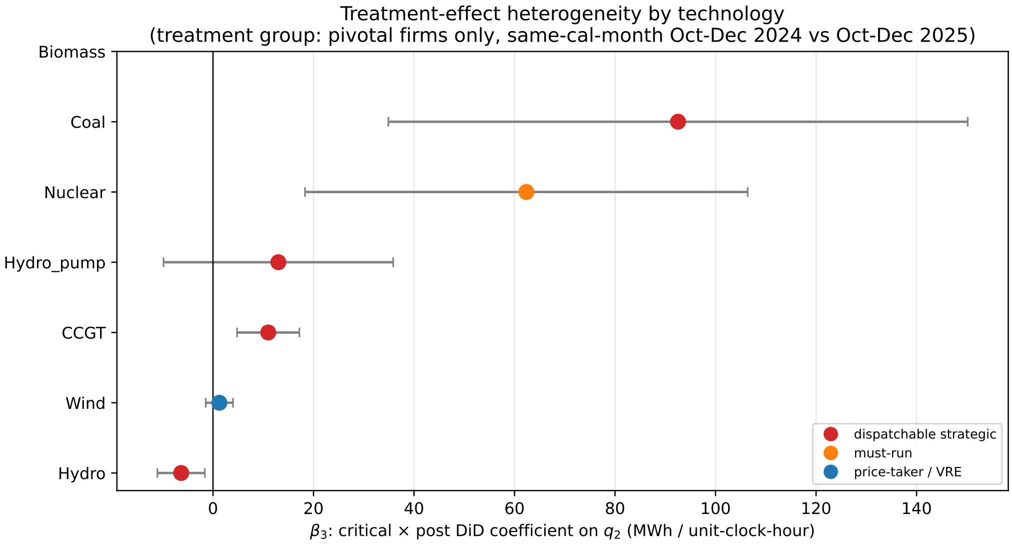

Figure 5: B1 – Treatment-effect heterogeneity by technology. Coefficient plot of β_3 (critical × post DiD interaction) on q_2, stratified by `tech_group` within the treatment group. Error bars are 95% confidence intervals using cluster-robust standard errors by date. Color coding: red = dispatchable strategic technologies (CCGT, Hydro, Hydro_pump, Coal); orange = must-run (Nuclear); blue = price-taker / VRE (Wind). The dispatchable–non-dispatchable split is the key validation of the within-day-DiD design: granularity has economic content for technologies that can strategically reshape supply, not for must-run or price-taker technologies. Source: Table 3.

Table 3: B1 – β_3 stratified by technology, treatment vs placebo group. Same specification as Table 2, restricted by technology and group. The "Predicted" column shows the model's directional prediction: + for dispatchable strategic technologies (CCGT, Coal, Hydro, Hydro_pump), 0 for must-run (Nuclear) and price-taker (Wind, Biomass). The treatment-group dispatchable techs all show large positive β_3; Wind shows the predicted null. The placebo group should show $\beta_3 \approx 0$ across all technologies. Significance markers as in Table 2.

	Treatment group		Placebo group		
Technology	β_3	SE / n	β_3	SE / n	Predicted
CCGT	10.99***	(3.17) / 63,708	17.51	(17.58) / 13,761	+
Hydro	−6.33**	(2.41) / 38,897	−0.08	(5.02) / 5,274	+
Hydro_pump	13.01	(11.66) / 6,706	—	—	+
Coal	92.54**	(29.41) / 806	—	—	+
Nuclear	62.35**	(22.47) / 20,632	—	—	0
Wind	1.27	(1.39) / 83,667	0.72***	(0.18) / 310,480	0
Biomass	—	—	—	—	0

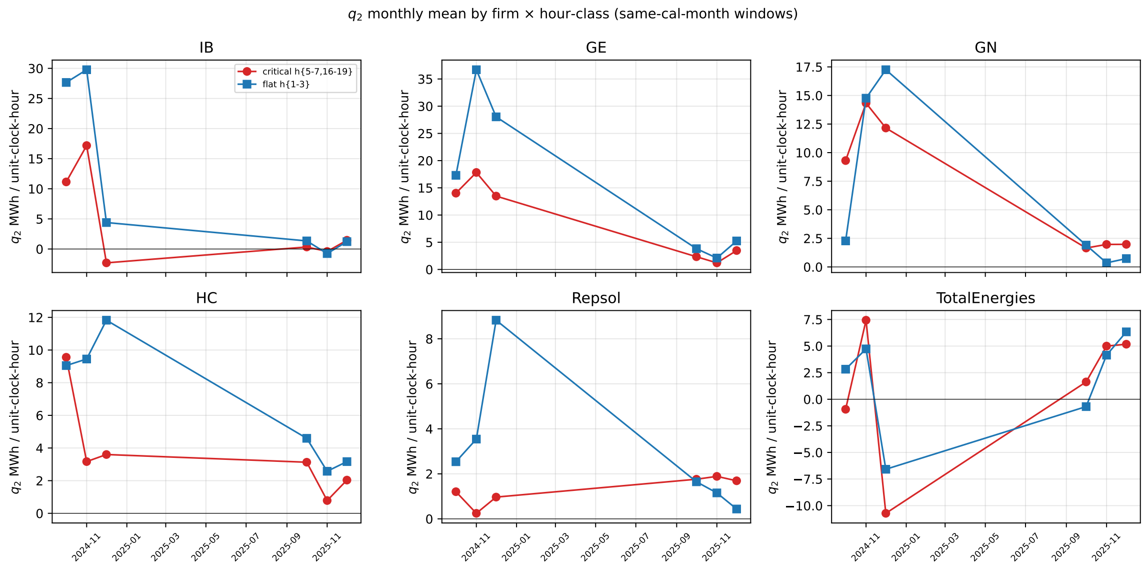

Figure 6: q_2 monthly mean by firm and hour-class, same-calendar-month windows. Six-panel monthly trajectory of q_2 (in MWh per unit-clock-hour) for the four treatment-group firms (Iberdrola, Endesa, Naturgy, EDP-Spain) and two placebo firms (Repsol, TotalEnergies). Red = critical hours $h\{18\text{-}22\}$; blue = flat hours $h\{3\text{-}5\}$. The two same-calendar-month windows (October–December 2024 and October–December 2025) bracket the MTU15-DA reform of October 1, 2025. Both windows are post-MTU15-IDA so the IDA reform of March 2025 is held constant. Source: `scripts/analysis/firm/thesis_figures.py`.

Table 4: Per-firm β_3 on q_2 (B1) and DA cleared MWh (B3). Joint per-firm view of the q_2 outcome (Ito–Reguant IDA upward adjustment) and DA cleared volume. The mechanism storyboard predicts *negative* β_3 on DA cleared (firms withhold in DA in critical hours post-reform) and *positive* β_3 on q_2 (firms add upward in IDA). EDP-PT has no `pibci` entries (Portuguese-zone units do not appear in OMIE intraday programmes); the DA-side effect is identified.

	q_2 outcome (B1)		DA cleared (B3)	
Firm	β_3	SE	β_3	SE
Iberdrola	11.00***	(2.89)	−34.72***	(10.13)
Endesa	11.15***	(2.19)	6.99*	(3.14)
Naturgy	0.87	(1.52)	−25.68***	(5.60)
EDP-Spain	3.37*	(1.55)	0.44	(2.44)
EDP-Portugal	—	—	−83.92***	(8.08)

5.2 DA $\rightarrow$ IDA price wedge: supporting outcome

Table 5: B2 – Within-day DiD on the DA $\rightarrow$ IDA price wedge. Outcome is hourly Spain DA clearing price minus IDA clearing price (averaged across IDA sessions) in EUR/MWh. Same-calendar-month restriction (Oct-Dec 2024 vs Oct-Dec 2025) shows the within-day differential expanded modestly post-reform. The full-window specification (2024-01 to 2025-12) attributes most of the within-day wedge spike to the post-blackout *operación reforzada* (April–September 2025): once the reforzada $\times$ critical interaction is included, the post-MTU15-DA $\times$ critical coefficient becomes large and negative, indicating that MTU15-DA itself *reduces* the within-day wedge after controlling for the regulatory overlay. The DA-IDA wedge is therefore a noisy outcome for MTU15-DA identification absent reforzada controls.

Specification	β_3	SE	p	G
Same-cal-month (Oct-Dec 2024 vs 2025)	4.62^{**}	(1.43)	0.0012	184
Full window, no controls	0.79	(1.18)	0.5043	730
Full window, + reforzada $\times$ crit	-10.06^{***}	(1.39)	0.0000	730
Full window, + reforzada + MTU15-IDA $\times$ crit	-10.06^{***}	(1.39)	0.0000	730

5.3 Other outcomes (DA cleared volume, bid-shape metrics)

5.4 Conditional parallel trends with renewable and reforzada controls

Table 6: B4 – Conditional parallel trends: spec stack for q_2, treatment group only. β_3 stability under successive control additions, starting from the baseline B1 specification (firm + DOW fixed effects). Spec (2) adds Spain hourly wind and solar production levels (ENTSO-E A75); Spec (3) adds the interaction of critical-hour dummy with VRE production (allowing the VRE effect to differ across critical and flat hours); Spec (4) adds calendar-month fixed effects. The same-calendar-month design already absorbs reforzada and MTU15-IDA via the post dummy (both hold constant within each window), so no separate controls for those reforms are needed in this spec stack. Stability of β_3 across columns is the conditional parallel-trends defence of the headline result.

Specification	β_3	SE	p	G
(1) Baseline (firm + DOW FE)	5.69^{***}	(1.08)	0.0000	184
(2) + wind, solar levels	5.65^{***}	(1.09)	0.0000	184
(3) + crit $\times$ VRE	5.41^{***}	(1.11)	0.0000	184
(4) + calendar-month FE	5.41^{***}	(1.11)	0.0000	184

6 Conclusion

A Robustness checks

A.1 Sensitivity to critical-hours definition

A.2 Sensitivity to firm-partition (pivotality vs administrative)

A.3 Same-cal-month vs full-window comparisons

Table 7: B5 – Robustness panel: β_3 across specifications, treatment group on q_2. Each row reports the headline coefficient β_3 under a single sensitivity. Section B5.1 varies the critical-hours definition between four candidates (price_peak is canonical). Section B5.2 varies the firm partition between the pivotality-based set (which includes EDP-PT) and the administrative IB/GE/GN/HC tier; the two coincide because EDP-PT lacks `pibci` data so its inclusion does not alter the q_2 regression. Section B5.3/B5.5 varies the window: the full 2024–2025 panel reduces the coefficient because the pre-MTU15-DA half of the panel is post-MTU15-IDA, where q_2 magnitudes are already crushed by the IDA reform; excluding the reforzada-active months (April–September 2025) recovers a coefficient close to the same-calendar-month estimate. Section B5.4 verifies that excluding individual units (EDP-PT, ABO2G) does not change the result.

Sensitivity	β_3	SE	p	G
B5.1 — Critical-hours definition				
supply_ramp h{7,8,16,17,18}	6.20***	(1.19)	0.0000	184
price_peak h{18-22}	8.20***	(1.49)	0.0000	184
demand_peak h{16-20}	7.85***	(1.34)	0.0000	184
joint h{7-8, 16-22}	6.98***	(1.27)	0.0000	184
B5.2 — Firm partition				
Pivotality-based treatment set	5.69***	(1.08)	0.0000	184
Administrative IB/GE/GN/HC	5.69***	(1.08)	0.0000	184
B5.3 / B5.5 — Window				
Full window 2024-2025	2.51***	(0.35)	0.0000	730
Full window minus Apr-Sep 2025	5.17***	(0.58)	0.0000	575
B5.4 — Sample exclusions				
Drop EDP-PT	5.69***	(1.08)	0.0000	184
Drop ABO2G	5.69***	(1.08)	0.0000	184
B5.6 — DST transition days (CET ↔ CEST)				
Same-cal-month, drop DST transition days	5.79***	(1.09)	0.0000	182
Full window, drop DST transition days	2.50***	(0.35)	0.0000	726
B5.7 — DST regime separation (clock-hour semantics differ across CEST/CET)				
Same-cal-month, CEST days only (Oct 1–25)	0.57	(1.11)	0.6077	50
Same-cal-month, CET days only (Oct 27–Dec 31)	7.57***	(1.35)	0.0000	132

A.4 Placebo specifications

A.5 Operational vs strategic decomposition (PIBCA vs PHF)

A.6 Bid-shape robustness in the competitive-zone restriction

B Data appendix

B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

B.2 File-format specifications and parser conventions

B.3 Power-vs-energy unit discipline

B.4 Coverage and missingness

B.5 Hour-of-day classification metrics

Table 8: Hour-of-day classification metrics, Spain 2025. For each clock-hour (Madrid local time), mean load, mean solar production, mean net-load (load − wind − solar), within-hour standard deviation of net-load σ_{within}, and mean signed within-hour load change $\Delta\text{load} = \text{load}_{q_4} - \text{load}_{q_1}$. Classification: **critical** hours have high σ_{within}; *flat* hours have low σ_{within}; *dropped* hours are borderline (moderate variance, no clean assignment).

Clock-hour	demand (GW)	solar (GW)	residual demand (GW)	σ_{within} (MW)	within-hour Δ dema
00:00–01:00	23.2	0.2	16.1	398	
01:00–02:00	22.4	0.1	15.4	356	
02:00–03:00	22.0	0.1	15.1	272	
03:00–04:00	21.9	0.1	15.2	328	
04:00–05:00	22.9	0.1	16.3	654	
05:00–06:00	24.9	0.3	18.4	768	
06:00–07:00	27.0	1.9	18.9	1004	
07:00–08:00	28.3	5.8	16.7	1261	
08:00–09:00	28.7	10.5	12.9	1122	
09:00–10:00	28.8	13.5	10.2	795	
10:00–11:00	28.6	15.0	8.7	583	
11:00–12:00	28.7	15.5	8.2	481	
12:00–13:00	28.6	15.5	8.0	438	
13:00–14:00	28.3	15.2	7.9	458	
14:00–15:00	28.0	14.4	8.2	550	
15:00–16:00	28.3	13.0	9.4	758	
16:00–17:00	28.9	10.1	12.5	1237	
17:00–18:00	30.0	6.4	16.7	1367	
18:00–19:00	31.1	2.6	21.3	1186	
19:00–20:00	32.0	0.7	23.8	549	
20:00–21:00	31.0	0.3	23.2	619	
21:00–22:00	28.6	0.2	20.9	801	
22:00–23:00	26.4	0.2	18.8	695	
23:00–00:00	24.6	0.2	17.2	536	

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